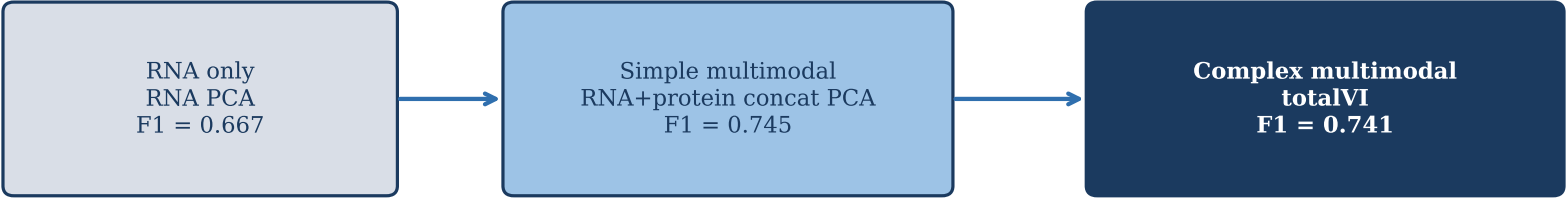


Figure 1. Protein-access gain vs residual totalVI-associated contrast

Protein-access gain
 $G_{\text{access}} = +0.078$
(strict train-only)

Residual totalVI-associated contrast
 $G_{\text{assoc}} = -0.004$
(near null; CI crosses 0)



Stress-test the residual totalVI-associated contrast under:
dataset transfer (inductive + feature-access-matched transductive) · protein degradation · training stochasticity · uncertainty diagnostics
External donor evidence (Lawlor): multimodal measurement value across 10 donors — not universal architecture superiority

Descriptive contrasts only (not causal architecture effects; primary = strict train-only 7,573/1,921):
 $G_{\text{access}} = F1_{\text{concat}} - F1_{\text{RNA_PCA}}$ · $G_{\text{assoc}} = F1_{\text{totalVI}} - F1_{\text{concat}}$
Central question: How much of the observed multimodal gain is associated with protein access,
and is there a stable residual totalVI-associated contrast beyond a simple RNA-protein representation?
(R_{access} is a secondary descriptive ratio only; not a recovery fraction.)